Assignment 1 Alvaro Méndez (2185566)

1 Section 1: Scottish Lakes

```
lake_names <- c("Loch Ness", "Loch Lomond", "Loch Morar", "Loch Tay",
                "Loch Awe", "Loch Maree", "Loch Ericht", "Loch Lochy",
                "Loch Rannoch", "Loch Shiel", "Loch Katrine", "Loch Arkaig"
                    ,
                "Loch Shin")
volume <- c(7.45, 2.6, 2.3, 1.6, 1.2, 1.09, 1.08, 1.07, 0.97, 0.79, 0.77,
    0.75, 0.35)

# Area in km^2
area <- c(56, 71, 27, 26.4, 39, 28.6, 18.6, 16, 19, 19.5, 12.4, 16, 22.5)

# Length in km
length_km <- c(39, 36, 18.8, 23, 41, 20, 23, 16, 15.7, 28, 12.9, 19.3,
    27.8)

# Maximum depth in m
max_depth <- c(230, 190, 310, 150, 94, 114, 156, 162, 134, 128, 151, 109,
    49)

# Mean depth in m
mean_depth <- c(132, 37, 87, 60.6, 32, 38, 57.6, 70, 51, 40, 43.4, 46.5,
    15.5)


# Create a data frame
scottish.lakes <- data.frame(Name = lake_names, Volume_km3 = volume, Area_
    km2 = area,
                        Length_km = length_km, Max_Depth_m = max_depth,
                            Mean_Depth_m = mean_depth)
str(scottish.lakes)

max_volume <- scottish.lakes$Volume_km3[which.max(scottish.lakes$Volume_km3
    )]
max_volume_lake <- scottish.lakes$Name[which.max(scottish.lakes$Volume_km3)
    ]
cat("The lake with the largest volume is", max_volume_lake, "with a volume
    of", max_volume, "km^3.\n")
min_volume <- scottish.lakes$Volume_km3[which.min(scottish.lakes$Volume_km3
    )]
min_volume_lake <- scottish.lakes$Name[which.min(scottish.lakes$Volume_km3)
    ]
cat("The lake with the smallest volume is", min_volume_lake, "with a volume
     of", min_volume, "km^3.\n")
```

```r
max_area <- scottish.lakes$Area_km2[which.max(scottish.lakes$Area_km2)]
max_area_lake <- scottish.lakes$Name[which.max(scottish.lakes$Area_km2)]
cat("The lake with the largest area is", max_area_lake, "with an area of",
    max_area, "km^2.\n")
min_area <- scottish.lakes$Area_km2[which.min(scottish.lakes$Area_km2)]
min_area_lake <- scottish.lakes$Name[which.min(scottish.lakes$Area_km2)]
cat("The lake with the smallest area is", min_area_lake, "with an area of",
     min_area, "km^2.\n")


scottish.lakes_ord <- scottish.lakes[order(scottish.lakes$Area_km2), ]

largest_two <- tail(scottish.lakes_ord, 2)[, c("Name", "Area_km2")]
cat("The two lakes with the largest area are: \n")
largest_two

tot_area <- sum(scottish.lakes_ord$Area_km2)
cat("The total area of the ", length(scottish.lakes$Name)," lakes is", tot_
    area, "km^2).\n")
```

The output of the above code is as follows:

```
  The lake with the largest volume is Loch Ness with a volume of 7.45 km^3.
The lake with the smallest volume is Loch Shin with a volume of 0.35 km^3.
The lake with the largest area is Loch Lomond with an area of 71 km^2.
The lake with the smallest area is Loch Katrine with an area of 12.4 km^2.
The two lakes with the largest area are:
         Name Area_km2
1   Loch Ness       56
2 Loch Lomond       71
The total area of the  13  lakes is 372 km^2).
```

2 Section 2: 100 m Athletics Records

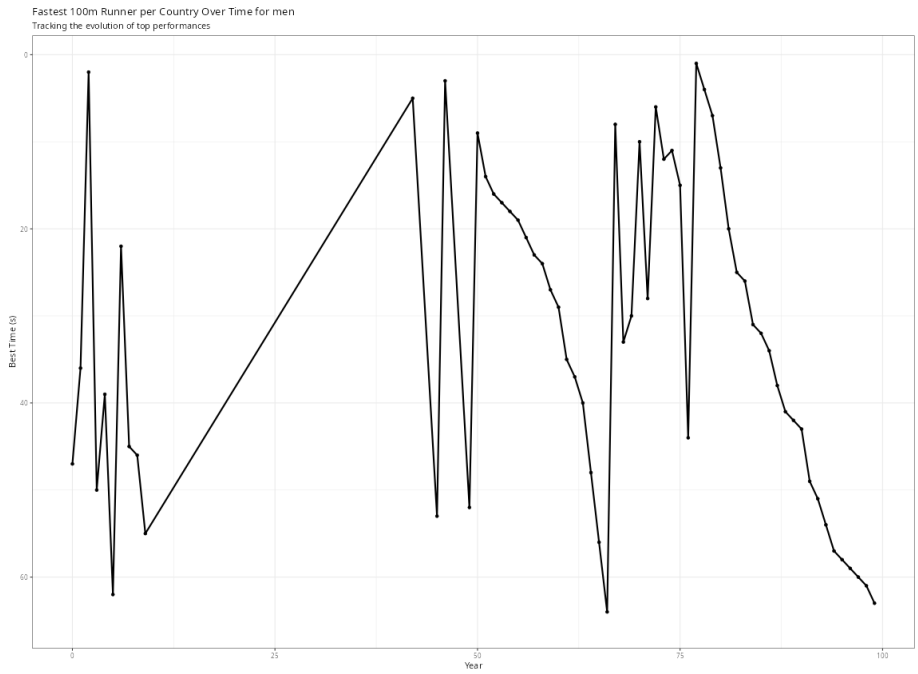
2.1 Men's 100m

```r
library(rvest)
library(tidyverse)
library(ggplot2)
# For the men's 100m records, we can scrape the data from the All-Time
    Athletics website.
men100m_html <- read_html("https://www.alltime-athletics.com/m_100ok.htm")
men100m_html |> html_nodes(xpath = "//pre") |> html_text() -> men100m_list
men100m_tbl <- read_fwf(men100m_list)
str(men100m_tbl)

men100m_tbl <- men100m_tbl %>%
  mutate(across(c(X2, X3,X7), as.numeric),X9 = as.Date(X9, format = "%d.%m
      .%Y"), X6 = as.Date(X6, format = "%d.%m.%Y"))
```

```r
men100m_tbl <- men100m_tbl %>%
  rename(Rank = X1, Time = X2, Wind = X3, Athlete = X4, Country = X5, Venue
      = X7, Date = X6, Notes = X8, DOB = X9)

str(men100m_tbl)

# Find the best time for each country each year
fastest_per_country <- men100m_tbl %>%
  mutate(Year = year(Date)) %>%
  group_by(Country, Year) %>%
  summarise(Best_Time = min(Time), .groups = 'drop')

# Plot as a line chart
ggplot(fastest_per_country, aes(x = Date, y = order(Best_Time), color =
    Country)) +
  geom_line(size = 1) +
  geom_point() +
  scale_y_reverse() +
  theme_bw() +
  labs(title = "Fastest 100m Runner per Country Over Time",
       subtitle = "Tracking the evolution of top performances",
       y = "Best Time (s)")

men100m_tbl <- men100m_tbl %>%
  mutate(Year = year(Date), Decade_Label = paste0(floor(Year / 10) * 10, "s
     "))

ggplot(men100m_tbl, aes(x = fct_infreq(Country), fill = as.factor(Decade_
    Label))) +
  geom_bar() +
  coord_flip() +
  theme_minimal() +
  labs(title = "Top 100m Times by Country and Decade",
       x = "Country",
       y = "Number of Records",
       fill = "Decade")
```

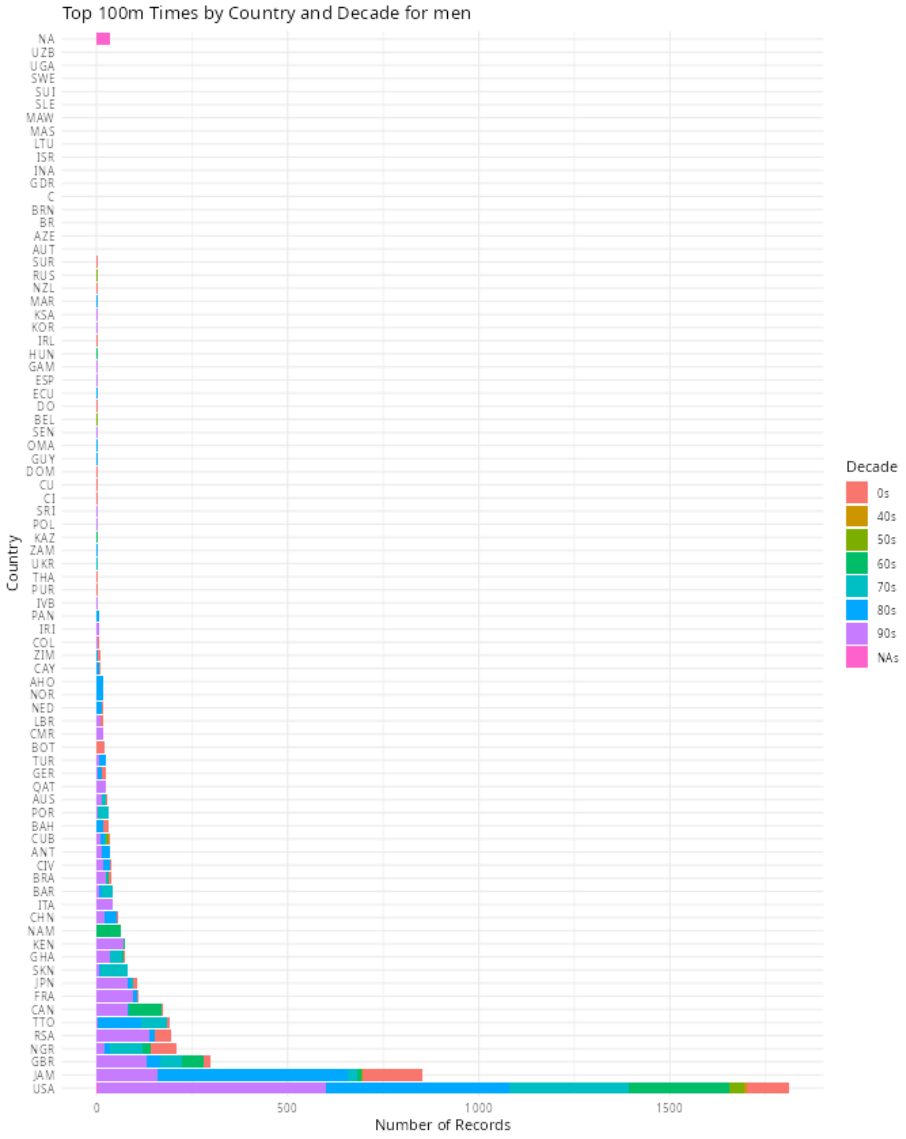
2.2 Women's 100m

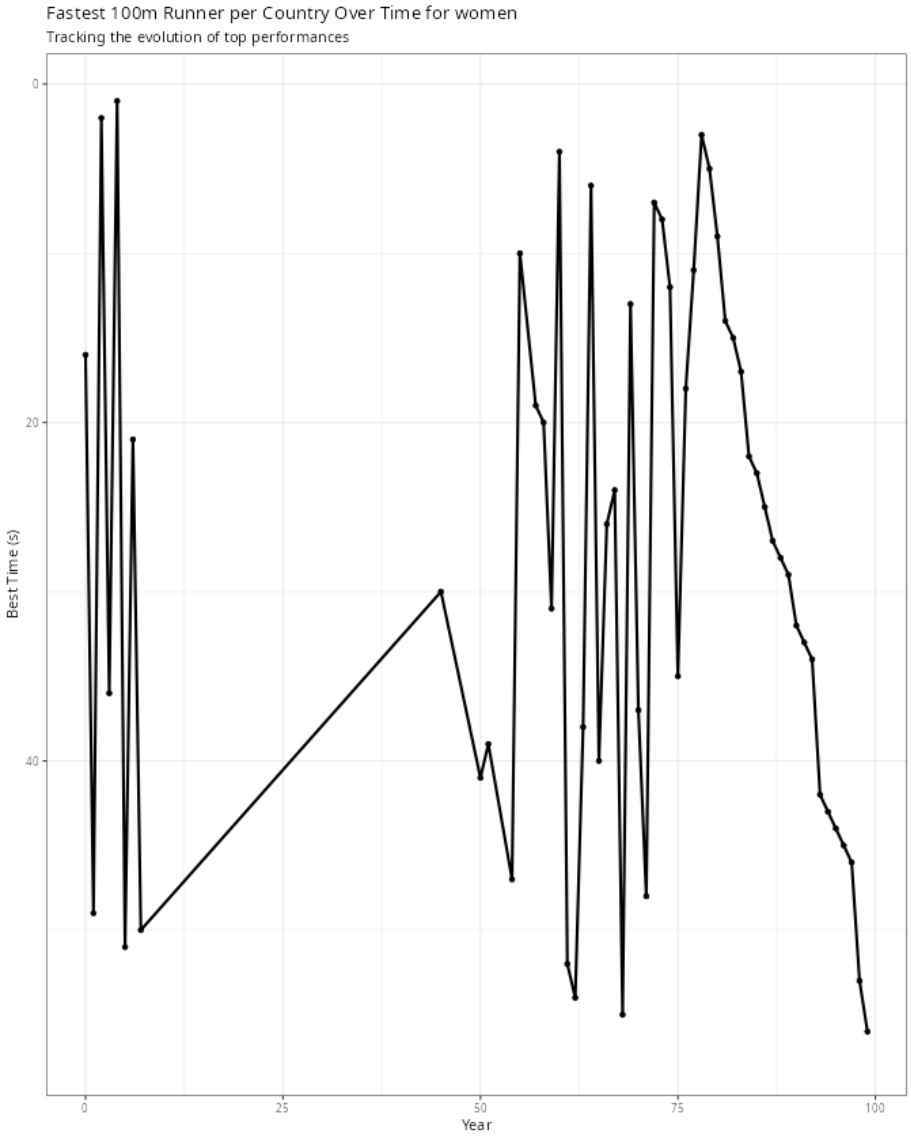
```r
# No we do the same for the women's 100m records.
women100m_html <- read_html("https://www.alltime-athletics.com/w_100ok.htm"
    )
women100m_html |> html_nodes(xpath = "//pre") |> html_text() -> women100m_
    list
women100m_tbl <- read_fwf(women100m_list)
str(women100m_tbl)


women100m_tbl <- women100m_tbl %>%
```

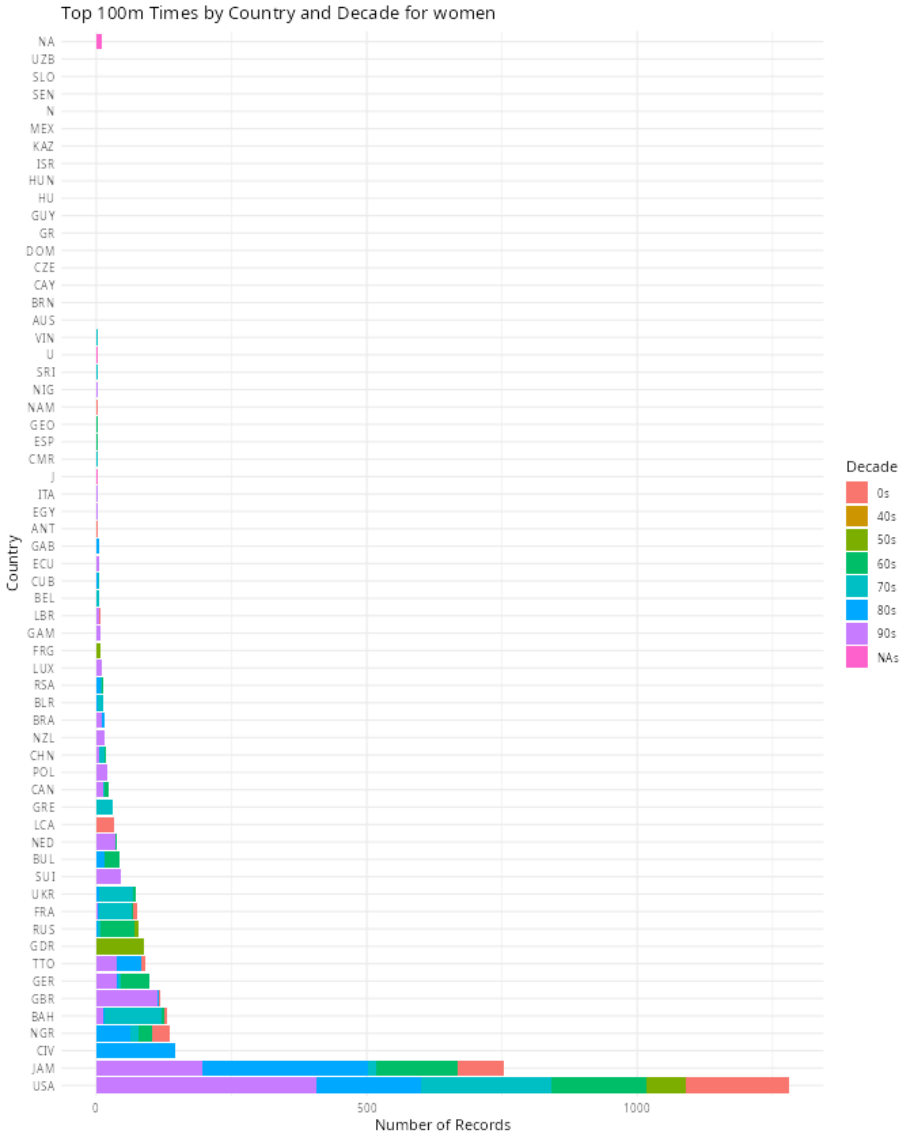



```
  mutate(across(c(X2, X3,X7), as.numeric),X9 = as.Date(X9, format = "%d.%m
      .%Y"), X6 = as.Date(X6, format = "%d.%m.%Y"))
women100m_tbl <- women100m_tbl %>%
  rename(Rank = X1, Time = X2, Wind = X3, Athlete = X4, Country = X5, Venue
       = X7, Date = X6, Notes = X8, DOB = X9)

str(women100m_tbl)

# Find the best time for each country each year
fastest_per_country <- women100m_tbl %>%
  mutate(Year = year(Date)) %>%
  group_by(Year) %>%
  summarise(Best_Time = min(Time), .groups = 'drop')

# Plot as a line chart
ggplot(fastest_per_country, aes(x = Year, y = order(Best_Time))) + #, color
     = Country
  geom_line(size = 1) +
  geom_point() +
  scale_y_reverse() +
  theme_bw() +
  labs(title = "Fastest 100m Runner per Country Over Time for women",
       subtitle = "Tracking the evolution of top performances",
       y = "Best Time (s)")
```

```r
women100m_tbl <- women100m_tbl %>%
  mutate(Year = year(Date), Decade_Label = paste0(floor(Year / 10) * 10, "s
      "))

ggplot(women100m_tbl, aes(x = fct_infreq(Country), fill = as.factor(Decade_
    Label))) +
  geom_bar() +
  coord_flip() +
  theme_minimal() +
  labs(title = "Top 100m Times by Country and Decade for women",
       x = "Country",
       y = "Number of Records",
       fill = "Decade")
```


Top 100m Times by Country and Decade for men
Country
NA
UZB
UGA
SWE
SUI
SLE
MAW
MAS
LTU
ISR
INA
GDR
C
BRN
BR
AZE
AUT
SUR
RUS
NZL
MAR
KSA
KOR
IRL
HUN
GAM
ESP
ECU
DO
BEL
SEN
OMA
GUY
DOM
CU
CI
SRI
POL
KAZ
ZAM
UKR
THA
PUR
IVB
PAN
IRI
COL
ZIM
CAY
AHO
NOR
NED
LBR
CMR
BOT
TUR
GER
QAT
AUS
POR
BAH
CUB
ANT
CIV
BRA
BAR
ITA
CHN
NAM
KEN
GHA
SKN
JPN
FRA
CAN
TTO
RSA
NGR
GBR
JAM
USA
0
500
1000
1500
Number of Records
Decade
0s
40s
50s
60s
70s
80s
90s
NAs


Fastest 100m Runner per Country Over Time for women
Tracking the evolution of top performances
Best Time (s)
0
20
40
0
25
50
75
100
Year


Top 100m Times by Country and Decade for women
Country
NA
UZB
SLO
SEN
N
MEX
KAZ
ISR
HUN
HU
GUY
GR
DOM
CZE
CAY
BRN
AUS
VIN
U
SRI
NIG
NAM
GEO
ESP
CMR
J
ITA
EGY
ANT
GAB
ECU
CUB
BEL
LBR
GAM
FRG
LUX
RSA
BLR
BRA
NZL
CHN
POL
CAN
GRE
LCA
NED
BUL
SUI
UKR
FRA
RUS
GDR
TTO
GER
GBR
BAH
NGR
CIV
JAM
USA
0
500
1000
Number of Records
Decade
0s
40s
50s
60s
70s
80s
90s
NAs